Project Planning Phase

Project Planning (Product Backlog, Sprint Planning, Stories, Story points)

Date	07 March 2026
Team ID	**Team 295**
Project Name	UCab – Cab Booking Platform
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation:

Here's a completed Product Backlog, Sprint Schedule, and Estimation table for our **UCab – Cab Booking Platform**, developed using the **MERN stack (MongoDB, Express.js, React.js, Node.js)**. The backlog includes epics such as user authentication, driver management, ride booking, ride matching, real-time tracking, notifications, and trip history. These features are divided into user stories and distributed across multiple sprints for systematic development and delivery:

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a passenger, I can register using email and password	2	High	1,2,3,4
Sprint-1	Registration	USN-2	As a driver, I can create an account and upload basic details	2	High	1,2,3,4
Sprint-1	Login	USN-3	As a user, I can login securely to the system	2	High	1,2,3,4
Sprint-1	Authentication	USN-4	As a system, I implement JWT authentication for secure sessions	3	High	1,2,3,4
Sprint-2	Ride Booking	USN-5	As a passenger, I can request a ride by selecting pickup and drop location	5	High	1,2,3,4
Sprint-2	Driver Availability	USN-6	As a driver, I can set my availability status (online/offline)	3	High	1,2,3,4
Sprint-2	Ride Matching	USN-7	As a system, I match passengers with nearby drivers	5	High	1,2,3,4
Sprint-3	Ride Management	USN-8	As a driver, I can accept or reject ride requests	3	High	1,2,3,4

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Trip Tracking	USN-9	As a passenger, I can track the driver location in real time	5	High	1,2,3,4
Sprint-4	Trip History	USN-10	As a user, I can view my completed rides history	3	Medium	1,2,3,4
Sprint-4	Notifications	USN-11	As a user, I receive ride status notifications	2	Medium	1,2,3,4
Sprint-4	Security	USN-12	As a system, I implement secure APIs and data protection	2	High	1,2,3,4

Project Tracker, Velocity & Burndown Chart:

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	18	6 Days	05 Feb 2026	09 Feb 2026	18	09 Feb 2026
Sprint-2	13	6 Days	10 Feb 2026	14 Feb 2026	-	
Sprint-3	8	6 Days	15 Feb 2026	19 Feb 2026	-	
Sprint-4	7	6 Days	20 Feb 2026	24 Feb 2026	-	

x

Velocity:

Imagine we have a 6-day sprint duration, and the velocity of the team is 18 (points per sprint). Let’s calculate the team’s average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{18}{1} = 18$$

Burndown Chart:

Here's a simple Burndown Chart representation for the **UCab – Cab Booking Platform**, based on the planned **4-sprint Agile development cycle**. The chart shows the remaining story points after each sprint, helping the team track development progress and ensure that the planned features such as ride booking, driver matching, real-time tracking, and trip management are completed within the project timeline:

sprint	planned points	remaining points
0	_	46
1	18	28
2	13	15
3	8	7
4	7	0

Burndown Chart Explanation

- X-axis: Sprint Number (0–4)
- Y-axis: Remaining Story Points (46 → 0)
- Ideal Line: Straight diagonal from (0,46) to (4,0)
- Actual Line: Shows real sprint progress

If the actual line stays below the ideal line, the team is ahead of schedule